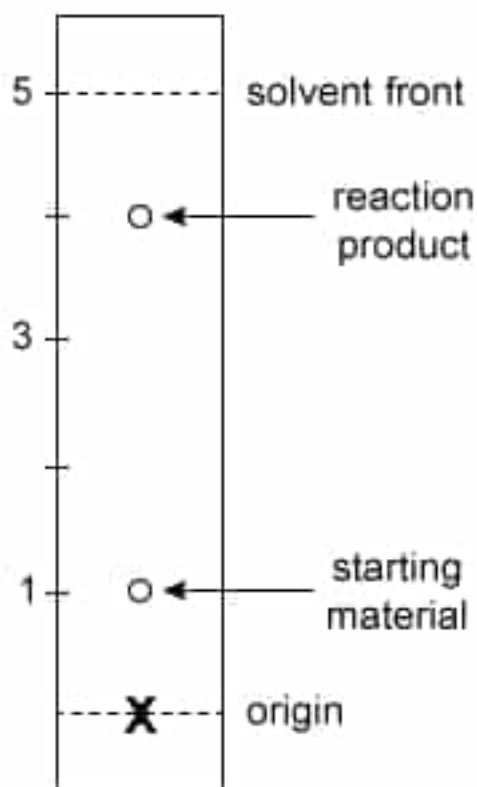


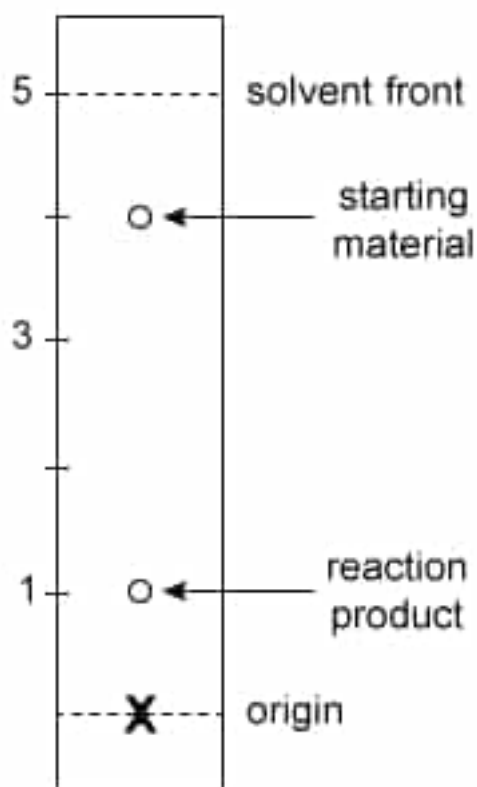
A researcher uses thin-layer chromatography (TLC) to monitor a reaction where the starting material has an R_f of 0.8 and the reaction product has an R_f of 0.2. Which TLC plate correctly depicts the positions of the starting material and reaction product?

☐ A.



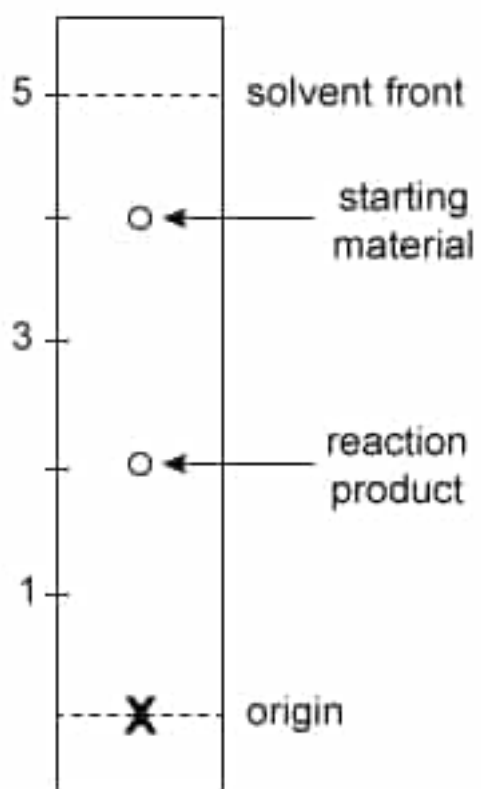
[10%]

✓ ☒ B.



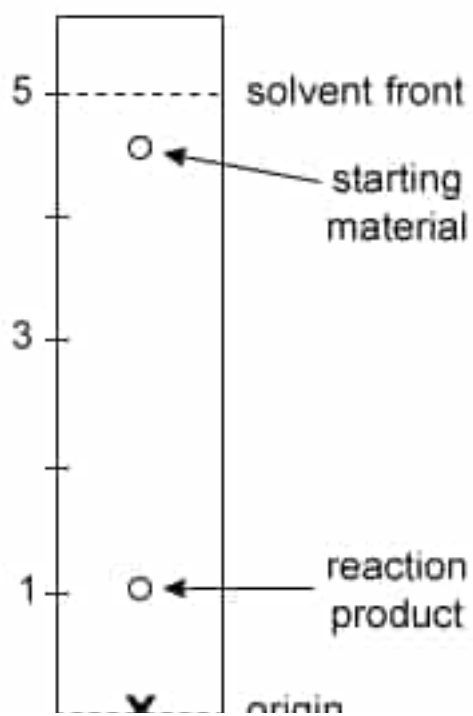
[79%]

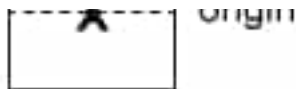
☐ C.



[3%]

☐ D.





[5%]

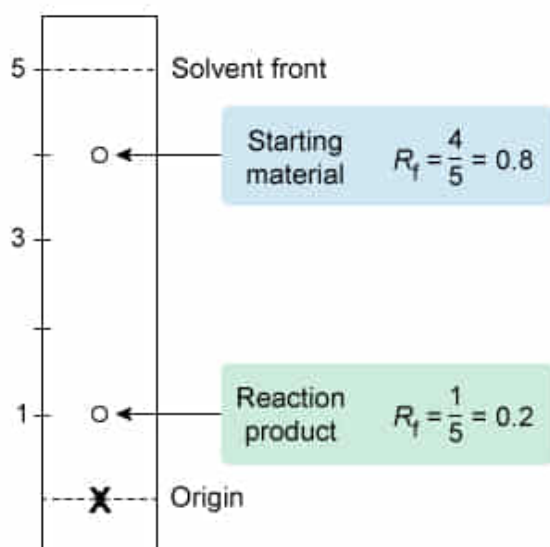
Correct

79% answered correctly

Explanation:

Thin-layer chromatography and R_f values

$$R_f = \frac{\text{Distance compound traveled}}{\text{Distance solvent traveled}}$$



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Thin-layer chromatography (TLC) is a technique used to separate mixture components based on polarity. The TLC plate is placed in a solvent chamber after a small amount of a mixture is spotted on the origin of the TLC plate. The solvent (mobile phase) travels up the TLC

plate (stationary phase) via **capillary action**.

The mixture components separate because they travel up the TLC plate at different rates, depending on the strength of competing **intermolecular interactions** between the mixture component and the stationary and mobile phases. The **R_f value** is a unitless ratio that describes how far a compound travels up the TLC plate compared to how far the solvent traveled, and is calculated using:

$$R_f = \frac{\text{Distance compound traveled from origin}}{\text{Distance from origin to solvent front}}$$

R_f values can range from 0 to 1, and the greater the R_f value, the farther the compound has traveled up the plate.

The question states that the starting material for a reaction has an R_f of 0.8 and the reaction product has an R_f of 0.2. The TLC plates show that the distance from the origin to the solvent front is 5. Using these values and the equation above:

$$0.8 = \frac{\text{Distance starting material traveled from origin}}{5}$$

$$\text{Distance starting material traveled from origin} = 0.8 \times 5 = 4$$

$$0.2 = \frac{\text{Distance reaction product traveled from origin}}{5}$$

$$\text{Distance starting material traveled from origin} = 0.2 \times 5 = 1$$

Choice B is the only TLC plate that shows the starting material with an R_f of 0.8 (having traveled a distance of 4) and the reaction product with an R_f of 0.2 (having traveled a distance of 1).

(Choice A) The positions of the starting material and the reaction product are inverted on this TLC plate.

(Choice C) This TLC plate incorrectly shows the reaction product having traveled a distance of 2, which gives an R_f value of 0.4 ($R_f = 2/5$

= 0.4) instead of 0.2.

(Choice D) This TLC plate incorrectly shows the starting material having traveled a distance of 4.5, which gives an R_f value of 0.9 ($R_f = 4.5/5 = 0.9$) instead of 0.8.

Educational objective:

Thin-layer chromatography (TLC) is a technique used to separate mixture components based on polarity. The R_f value is a unitless ratio of how far a compound traveled up the TLC plate compared to how far the solvent traveled.

Time spent:0 sec

Subject:
Organic Chemistry

QID:404123

System:
Separation Techniques, Spectroscopy, and
Analytical Methods